Debjani Nath Sarma

Lawrence Township, New Jersey, 08648 | debjanisarma7@gmail.com | +1 (609) 302-0587 | LinkedIn | GitHub |Portfolio

PROFESSIONAL SUMMARY

Data Analyst with experience spanning multiple industry verticals, including banking, retail, energy and pharmaceutical domains. Skilled in SQL, Python and Power BI to translate complex requirements into data-driven solutions and actionable insights. Proven ability to bridge requirements into technical specifications across ERP, CRM, and API integration environments.

EDUCATION

Master of Science in Business Analytics **December 2025**
University of South Florida, Tampa, Florida (3.97/4.00)

Bachelor of Technology in Computer Science and Engineering **July 2015**
Sikkim Manipal University, India

SKILLS

Data and Analytics: SQL (SSMS, MySQL, Oracle DB), Python (Pandas, NumPy, Scikit-learn, Matplotlib), R, Advanced Excel

BI and Visualization: Power BI (DAX, Power Query), Tableau, Data Storytelling, Dashboarding

Data Engineering: ETL/ELT Pipelines, OLAP, Data Warehousing, Data Cleaning / Wrangling, REST/SOAP APIs, System Integration

Modeling and Statistics : Predictive Modeling, Random Forest, XGBoost, Linear/ Logistic Regression, Classification, Clustering (K-Means), A/B Testing, Hypothesis Testing, SHAP

Business Analysis: Requirements Gathering (BRD/FRD), Root Cause Analysis, UAT Coordination, Process Improvement, Stakeholder Management, Data Flow Diagrams, Data Storytelling, Agile/Scrum, JIRA, Git / GitHub

PROFESSIONAL EXPERIENCE

Research Assistant | University of South Florida | Tampa, Florida **February 2026 - Present**

- Processed 7M+ records from HCUP National Inpatient Sample using chunked loading and pandas/Python; built a three-model ML framework (XGBoost, Logistic Regression) with GridSearchCV tuning across 108 combinations, 5-fold stratified cross-validation, and class imbalance handling via loss-function reweighting.
- Applied SHAP (XAI) to reveal operational and clinical features outweigh financial metrics; designed a leave-one-dimension-out convergence test confirming empirical dimensional independence across a 9-variable construct.

Graduate Teaching Assistant | University of South Florida | Tampa, Florida **August 2025 - December 2025**

- Instructed and mentored 130+ students in SQL, Python, Tableau, and Power BI across multiple courses, reinforcing best practices on data analysis, visualization, and business reporting.
- Held regular office hours to support students with coursework, debugging, and analytical problem-solving, translating complex technical concepts into clear and actionable guidance.

Graduate Data Analyst | University of South Florida | Tampa, Florida **May 2025 - July 2025**

- Integrated and analyzed FBI UCR and U.S. Census (2010-2019) across 50 states, performing data cleaning, normalization, and feature engineering in Python to enable consistent cross-state and temporal analysis.
- Built interactive Tableau dashboards to evaluate relationships between socioeconomic factors (income, unemployment, demographics) and crime trends; identified patterns such as higher property crime in low-income states and weak correlation between unemployment and violent crime, enabling data-driven policy insights.

Senior Data Analyst | Accenture | Bengaluru, India **January 2022 - May 2023**

- Collaborated with cross-functional stakeholders for a global fashion retailer to architect a unified analytics data flow, improving consistency across 3+ downstream reporting systems.
- Analyzed historical sales and product performance to identify top-selling categories and seasonal trends, helping optimize inventory planning and reducing stockouts by 15%.
- Led root cause analysis of data discrepancies between POS and inventory systems, improving data integrity and reporting reliability by ~20%.
- Conducted UAT and reporting validation, ensuring analytics solutions aligned with business requirements and decreased reporting defects by 25%.

Consultant | Software AG | Bengaluru, India **March 2020 - December 2021**

- Developed RESTful API pipelines to streamline financial transaction data for real-time compliance reporting and auditing at a Tier-1 global bank.

- Led a team of 4 engineers to deliver robust integration modules on schedule, coordinating sprint planning, task allocation and progress tracking using JIRA to achieve key project milestones.
- Drove cross-functional UAT and go-live coordination with compliance stakeholders, ensuring full test coverage and defect resolution, resulting in real-time transaction visibility and a centralized audit trail.

Junior Analyst | Accenture | Bengaluru, India **March 2016 - February 2020**

- Built enterprise integrations across SAP HANA, IBM Maximo, Ariba and CRM systems using webMethods ESB across multiple projects for clients from entertainment, energy, pharmaceutical and utilities industries.
- Implemented and tested REST & SOAP APIs in webMethods, to streamline data pipelines and improve interoperability.
- Gathered and translated business requirements into technical data flow specifications and integration designs, reducing ambiguity and rework during development cycles.

PROJECT EXPERIENCE

Movie Recommendation Strategy (Oracle SQL, Tableau, Python) **June 2025**

- Designed a multi-dimensional data warehouse and built OLAP schemas for 45,000+ movies and 20M+ user ratings to perform SQL analysis based on users, time-period, region, genre ROI, audience sentiment, and release trends.
- Developed interactive Tableau dashboards for data storytelling, visualizing genre performance, ROI, and budget-revenue relationships; integrated insights with Random Forest modeling to identify high-ROI genres (Crime, Comedy) and support data-driven film strategy decisions.

Airline Customer Retention (Python, Predictive Modeling) **January 2025**

- Performed feature engineering, data preprocessing and exploratory data analysis on an airline customer survey dataset with 100,000+ records using Pandas, NumPy and Matplotlib to detect patterns and trends.
- Designed Random Forest model with 96% accuracy and an XG Boost model with 95% accuracy using Python to identify key factors influencing customer retention in the airline industry.